

Photoionization rates per target atom at the end of reionization, in units directly comparable to a cosmic-ray ionization rate

Prepared for L. Carvalho

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Abstract

Converts the ionizing output of the reionization-era source population into Γ_i , the photoionization rate *per target particle* in s^{-1} , for H I, He I and He II — the same units as a cosmic-ray ionization rate ζ . The hydrogen rate is taken from direct measurement; the helium rates are derived as ratios to it, which cancels the background normalisation and leaves only the spectral shape and the photoionization cross-sections. Three spectral bases are compared. Script: `ionizing_rates_vs_cosmic_rays.py`.

1 Assumptions

- Redshift and phase.** All rates are quoted at $z = 6.0$ and $z = 4.9$, i.e. at and just after the end of reionization, where Γ_{HI} is measured. They apply to gas *inside ionized regions*; see Section 6.
- Escaping, not intrinsic.** The background is the one that survives escape from galaxies and propagation through the IGM, because that is what [?] measure from the Ly α forest.
- Optically thin targets.** Γ_i is computed for an unshielded atom bathed in the mean background. This is a good approximation for H I and He I at $z \simeq 6$ but *not* for He II (Section 6).
- Spectral shape.** Three bases, none of which is assumed correct a priori:
 - a power law $J_\nu \propto \nu^{-\alpha_s}$ with $\alpha_s = 2.0 \pm 0.6$, the effective ionizing-source index measured by [?];
 - the BPASS stellar SED of [?], Fig. 1, read off as a piecewise power law with indices 0.97 (13.6–24.6 eV), 2.18 (24.6–54.4 eV) and 23.8 above the He II edge. [Read off a published figure; I estimate the uncertainty at ± 0.2 dex per anchor point, which propagates to roughly ± 0.3 in each index.]
 - the Pop III photon budgets of [?], Table 4, used for photon-number ratios only.
- Cross-sections** are the standard analytic fits of Verner et al. (1996) / Osterbrock & Ferland (2006): $\sigma_{\text{HI}}(\nu_0) = 6.30 \times 10^{-18}$, $\sigma_{\text{HeI}}(\nu_0) = 7.42 \times 10^{-18}$, $\sigma_{\text{HeII}}(\nu_0) = 1.58 \times 10^{-18} \text{ cm}^2$. [These are textbook values, not from the 57-paper corpus.]
- Thresholds:** 13.598, 24.587, 54.418 eV (NIST). **Cosmology:** $\Omega_b h^2 = 0.02237$, $h = 0.6736$, $Y_p = 0.2454$ [?].

2 Derivation

The photoionization rate per target particle of species i is

$$\Gamma_i = \int_{\nu_i}^{\infty} \frac{4\pi J_\nu}{h\nu} \sigma_i(\nu) d\nu, \quad (1)$$

with J_ν the mean specific intensity. Writing $J_\nu = J_0 f(\nu)$ with $f(\nu_{\text{HI}}) \equiv 1$, the normalisation J_0 cancels in a ratio:

$$\frac{\Gamma_i}{\Gamma_{\text{HI}}} = \frac{\int_{\nu_i}^{\infty} f(\nu) \sigma_i(\nu) \frac{d\nu}{\nu}}{\int_{\nu_{\text{HI}}}^{\infty} f(\nu) \sigma_{\text{HI}}(\nu) \frac{d\nu}{\nu}}. \quad (2)$$

This is the whole point of the method: everything uncertain about the *amplitude* of the background drops out, and the answer depends only on the shape f and on atomic physics. Absolute rates then follow from the measured Γ_{HI} .

For the special case $f \propto \nu^{-\alpha}$ and $\sigma \propto \nu^{-3}$, Eq. 1 integrates in closed form to

$$\Gamma_i = \frac{4\pi \sigma_i(\nu_i) J_\nu(\nu_i)}{h(\alpha + 3)}, \quad (3)$$

which is used only as a sanity limit; the numbers in Section 4 use the full cross-sections in Eq. 2.

3 Verification

Symbolic. Eq. 3 is verified with `sympy` by direct integration (`simplify(Γsym-closed)=0`).

Dimensional. $[J_\nu] = \text{erg cm}^{-2} \text{s}^{-1} \text{Hz}^{-1} = E/L^2$, $[h\nu] = E$, $[\sigma] = L^2$, $[d\nu] = 1/T$, so $[\Gamma] = (E/L^2)(1/E)(L^2)(1/T) = 1/T$. Confirmed with `dimsys_SI.equivalent_dims`.

Independent numerical cross-check. Γ_{HI} is rebuilt from quantities that do *not* include it, using the local-source approximation $4\pi J_\nu = \epsilon_\nu^{\text{prop}} \lambda^{\text{prop}}$ with ϵ_{912} and λ_{mfp} from the same table [?]:

$$\Gamma_{\text{HI}}^{\text{rebuilt}} = 1.12 \times 10^{-13} \text{s}^{-1} \quad \text{vs} \quad \Gamma_{\text{HI}}^{\text{measured}} = 1.45 \times 10^{-13} \text{s}^{-1},$$

a ratio of 0.77. Agreement to 23% is as good as this approximation can be expected to do (it ignores the frequency dependence of the mean free path), and it confirms that the emissivity, mean free path and photoionization rate in that table are mutually consistent.

4 Results

Table 1: Photoionization rate ratios from Eq. 2. Independent of the background normalisation.

Spectral basis	$\Gamma_{\text{HeI}}/\Gamma_{\text{HI}}$	$\Gamma_{\text{HeII}}/\Gamma_{\text{HI}}$
(A) $\alpha_s = 1.4$ (-1σ)	0.706	3.60×10^{-2}
(A) $\alpha_s = 2.0$ (fiducial)	0.475	1.57×10^{-2}
(A) $\alpha_s = 2.6$ ($+1\sigma$)	0.322	6.81×10^{-3}
(B) BPASS stellar SED [?]	0.651	3.59×10^{-3}

Table 2: Absolute photoionization rates, s^{-1} per target particle. Γ_{HI} is measured [?]; helium rates follow from Table 1. Quoted Γ_{HI} uncertainties are the published total 1σ ; they scale the helium columns proportionally and are *not* the dominant error there — the spectral basis is.

z	basis	$\Gamma_{\text{HeI}} (\text{s}^{-1})$	$\Gamma_{\text{HeII}} (\text{s}^{-1})$
$\Gamma_{\text{HI}} = 1.45 (+1.57 / - 0.87) \times 10^{-13} \text{s}^{-1}$ at $z = 6.0$			
6.0	$\alpha_s = 1.4$	1.02×10^{-13}	5.22×10^{-15}
6.0	$\alpha_s = 2.0$	6.89×10^{-14}	2.27×10^{-15}
6.0	$\alpha_s = 2.6$	4.67×10^{-14}	9.88×10^{-16}
6.0	BPASS	9.44×10^{-14}	5.21×10^{-16}
$\Gamma_{\text{HI}} = 5.01 (+2.75 / - 2.32) \times 10^{-13} \text{s}^{-1}$ at $z = 4.9$			
4.9	$\alpha_s = 1.4$	3.54×10^{-13}	1.80×10^{-14}
4.9	$\alpha_s = 2.0$	2.38×10^{-13}	7.85×10^{-15}
4.9	$\alpha_s = 2.6$	1.62×10^{-13}	3.41×10^{-15}
4.9	BPASS	3.26×10^{-13}	1.80×10^{-15}

Photon-number rates. The literal “ionizing photons per unit time”, for completeness. At $z = 6.0$ [?], $\dot{n}_{\text{ion}} = 7.01 \times 10^{50} \text{s}^{-1} \text{cMpc}^{-3}$. With $\langle n_{\text{H}} \rangle = 1.895 \times 10^{-7} \text{cm}^{-3}$ comoving = $5.566 \times 10^{66} \text{cMpc}^{-3}$ this is $1.26 \times 10^{-16} \text{s}^{-1}$ per hydrogen atom, i.e. 3.97 ionizing photons per H atom per Gyr. The fraction emitted above each helium edge is 30.6% / 6.2% for basis (A) at $\alpha_s = 2.0$, and 32.5% / 0.63% for basis (B).

Pop III comparison (basis C). From [?], Table 4 (non-rotating, $\alpha_{\text{ov}} = 0.1$): $N_{\text{HeI}}/N_{\text{H}} = 0.384$ and $N_{\text{HeII}}/N_{\text{H}} = 0.0184$ for the top-heavy SB13 IMF, and 0.310 / 0.0096 for a Salpeter slope. A single power law cannot reproduce both ratios: they imply an effective index of 1.62 across the HeI edge but 2.88 across the HeII edge (SB13), confirming the curvature seen in basis (B). Metal-free stars are therefore harder than reionization-era populations in the HeI band but still show a HeII deficit.

5 Final result

At $z = 6$, per target particle, in the ionized IGM:

$$\begin{aligned}
 \Gamma_{\text{HI}} &= 1.45 (+1.57 / - 0.87) \times 10^{-13} \text{s}^{-1} && \text{(measured)} \\
 \Gamma_{\text{HeI}} &= (0.5\text{--}1.0) \times 10^{-13} \text{s}^{-1} && \text{i.e. } (0.32\text{--}0.71) \Gamma_{\text{HI}} \\
 \Gamma_{\text{HeII}} &\lesssim (0.5\text{--}5) \times 10^{-15} \text{s}^{-1} && \text{i.e. } (3.6 \times 10^{-3}\text{--}3.6 \times 10^{-2}) \Gamma_{\text{HI}}
 \end{aligned}$$

Γ_{HeI} is robust: every basis agrees to within a factor ~ 2 , because the 24.6–54.4 eV band is where stellar spectra are still strong and the IGM is transparent. Γ_{HeII} spans a factor ~ 10 and its lowest value — the BPASS one — is the physically appropriate one for a stellar population.

6 Caveats that matter for the cosmic-ray comparison

1. Γ is **per *neutral* target, and so is ζ** . The comparison Γ_i vs ζ_i is therefore apples-to-apples with no volume normalisation. If a volumetric rate is wanted instead, multiply by the density of the *neutral* species, not the total: inside ionized regions $x_{\text{HI}} \sim 10^{-4}\text{--}10^{-5}$, so $n_{\text{HI}}\Gamma_{\text{HI}} \ll n_{\text{H}}\Gamma_{\text{HI}}$.
2. Γ_{HeII} **is an upper bound**. HeII is not reionized at $z \simeq 6$ — it completes only by $z \sim 3$ [?] — so HeII is itself the dominant absorber above 54.4 eV and self-shields. The optically-thin

values in Table 2 therefore overestimate the true rate in the general IGM, by an amount that requires radiative transfer to quantify. [\[Not computed here.\]](#)

3. **These rates are zero in neutral regions.** Γ_{HI} is a volume-average over *ionized* gas. Outside ionization fronts the photoionizing background collapses, while cosmic rays and X-rays — with mean free paths far exceeding the bubble size — do not. This is the physically interesting regime for a cosmic-ray comparison: photons win by orders of magnitude inside H II regions and are essentially absent outside them, whereas cosmic-ray ionization is comparatively uniform [? ?].
4. **Helium abundance.** $n_{\text{He}}/n_{\text{H}} = 0.0813$ for $Y_{\text{p}} = 0.2454$, so even at equal Γ the helium *volumetric* ionization rate is $\sim 12\times$ smaller than hydrogen’s.

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